

A study of fault tolerance in star graph

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Abstract

The bounds on $f(n, k)$, the number of faulty nodes to make every $(n - k)$ -dimensional substar S_{n-k} in an n -dimensional star network S_n , have been derived. The exact value for $f(n, k)$ is determined when n is prime and $k = 2$, or when $n - 2 \leq k \leq n$. For $2 < k < n - 2$, a general method is presented to derive a set of faulty nodes which damage all S_{n-k} 's in S_n .

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1. Introduction

Since its discovery [1,2], much attention has been paid to the star graph interconnection networks as attractive alternatives to the popular hypercubes. Algebraic properties of star graphs were investigated in [1]. Star graph has superior node degree and diameter compared to the hypercube of a comparable size. More specifically, growth in node degree and diameter is sublogarithmic to the network size in star but logarithmic in hypercube. Various networks have been mapped to star graph [3]. A treatment of communication aspects for this network is presented in [4]. Fault tolerance of star graph has also been addressed extensively (see [5–7], for example).

The fault tolerance (resilience) of large networks is generally measured by how much of the network structure is preserved in the presence of a given number of node and/or link failures. Parallel algorithms run-

ning on the networks utilize the topological properties of these networks. Clearly, in the presence of node/link failures, the entire network is not available; thus the natural question is how large of a subnetwork (defined as a smaller network but with the same topological properties as the original one) is still available in the network in the presence of faults. In designing parallel processors using star graphs as the underlying interconnection topology as well as in designing real applications on such processors, the estimates of the network reliability and fault tolerance are important in choosing the algorithms and predicting their performance under failure conditions. Although, the first component failure (node or link) makes the complete star network unavailable, it is important to know how many fault-free star networks of smaller size (or substars) are available in the damaged structure. Under node-failure model, an enumeration of faulty nodes, which make every substar of a certain size faulty in a given star network, is presented here.

The rest of the paper is organized as follows. In Section 2, basic properties of the star graph and notations are presented. Section 3 forms the main body of the pa-

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per in which the main results are reported. Section 4 concludes the paper.

2. Preliminaries

The n -dimensional star graph, S_n , is an edge- and node-symmetric graph containing $n!$ nodes $(n-1)n!/2$ edges. The nodes are assigned labels each being a distinct permutation of the set of integers $\{1, 2, \dots, n\}$. Two nodes are joined with a link labeled i if and only if the label of one can be obtained from the label of the other by swapping the first digit (conventionally the leftmost) and the i th digit where $1 < i \leq n$. For instance, in a 5-star containing 120 nodes, two nodes 12345 and 42315 are neighbors and joined via an edge labeled 4. Each S_n contains n disjoint S_{n-1} 's. Partitioning an S_n can be done in $(n-1)$ ways. This is done by removing the set of i -links, $1 < i \leq n$, every time. For example, partitioning S_4 along the 4th link will result in $\{XXX4, XXX3, XXX2, XXX1\}$. Fig. 1 illustrates an S_4 where the mentioned substars are shown as hexagons.

It follows that the number of disjoint S_{n-k} 's in S_n is $k! \binom{n}{k}$ whereas the number of distinct S_{n-k} 's in S_n is $k! \binom{n-1}{k} \binom{n}{k}$. Let Λ_n be the symbol set $\{1, 2, \dots, n-1, n, X\}$, where X denotes a *don't care* symbol. Every substar of S_n can be uniquely labeled by a string of symbols in Λ_n such that the only repeated symbol be X . Notably, the number of X symbols in the label determines the dimension of the substar. For instance, the substar $X3X2X$ is an S_3 and contains the set of nodes $\{13425, 13524, 43125, 43521, 53124, 53421\}$. The first position (i.e., the leftmost position) of the label of any

substar is always equal to X because of connectivity conditions of the star graph, unless the substar is a single code (i.e., S_0). It is well known that the $n!$ permutations on the set $\{1, 2, \dots, n\}$ can be expressed as a sequence of transpositions of form (ij) . Accordingly, the nodes' labels in the star graph can be expressed as a sequence of odd or even transpositions wherein traversing one edge will change the node parity from odd to even (or from even to odd). This establishes that the star graph is bipartite. A k -tuple is a sequence $I_{m_1} I_{m_2} \dots I_{m_k}$ such that $1 < m_1 < m_2 < \dots < m_k \leq n$, and all I_{m_i} 's are distinct and belong to the set $\{1, 2, \dots, n\}$. If a given permutation on $n!$ nodes is such that it has digits $I_{m_1}, I_{m_2}, \dots, I_{m_k}$ in positions $m_1, m_2, \dots, m_k$, respectively, we say that the permutation *covers* the said k -tuple. For example, the 3-tuple $3_2 2_3 4_5$ is covered by the permutation 13254 or 53214.

3. Derivation of $f(n, k)$

We are interested in finding $f(n, k)$, the minimum number of nodes required to damage all S_{n-k} 's in S_n . Each node failure destroys $\binom{n-1}{k} S_{n-k}$'s in S_n . For example, the failure of node 12345 in S_5 will destroy $\binom{4}{2} = 6 S_3$'s, namely— $\{XXX45, XXX35, X2XX5, XXX34X, X2X4X, X23XX\}$.

Dividing the total no. of distinct S_{n-k} 's in S_n by the number of S_{n-k} 's destroyed by a single node will render: $f(n, k) \geq k! \binom{n}{k}$. This lower bound can be independently obtained by requiring one node failure for every disjoint S_{n-k} in $S_n (n!/(n-k)!)$. The upper bound is trivially obtained by using one node failure for each distinct S_{n-k} , i.e., $k! \binom{n-1}{k} \times \binom{n}{k}$. So it follows:

$$k! \binom{n}{k} \leq f(n, k) \leq k! \binom{n-1}{k} \binom{n}{k}.$$

As can be seen there is a large gap (factor of $\binom{n-1}{k}$) between the lower and upper bounds.

Example 1. $N = 5, p = 2, 20 \leq f(5, 2) \leq 120$.

Interestingly, the number of distinct S_3 's in S_5 is the same as the number of nodes in S_5 .

3.1. Special cases

Note that for some special cases the exact value for $f(n, k)$ coincides with the lower bound, and therefore is optimal. For example, $f(n, 0) = 1$ as the failure of a single node will fail the S_n . In addition, $f(n, 1) = n$, i.e., the exact number of node failures that destroy all

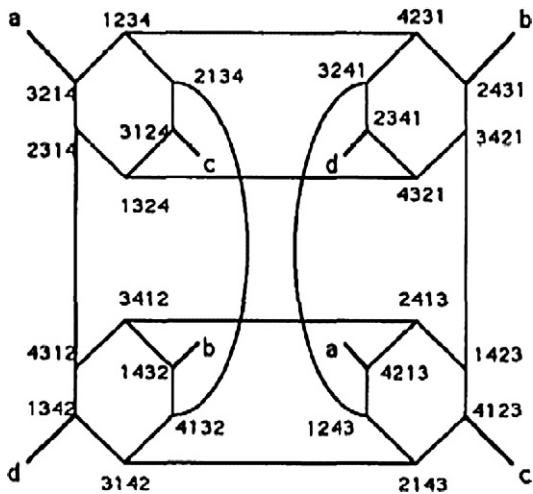


Fig. 1. The 4-dimensional star graph (S_4). Lines with the same labels are connected.

S_{n-1} 's is n . The labels of these nodes are $(123 \dots n)$ and its cyclic permutations.

For example, the set of faulty nodes $\{12345, 23451, 34512, 45123, 51234\}$ will destroy all S_4 's in S_5 . Furthermore, $f(n, n-2) = n!/2$. This number is obtained by simply having the set of faulty nodes that contain all nodes with even (or odd) permutations in their labels which optimally ensures the failure of every S_2 in S_n . Finally, it can be trivially shown that $f(n, n-1) = n!$.

3.2. Improving the bounds on $f(n, k)$

In this section, we derive results for $1 < k < n-2$. First, we consider the case where $k = 2$ and subsequently generalize the results to include other values of k .

Case (i). Derivation of $f(n, 2)$

To derive the minimum number of faulty nodes that can damage all S_{n-2} 's in S_n , we note that each S_{n-2} would span $(n-2)$ dimensions and thus would have as many X 's in its label. The remaining two positions (other than the leftmost position or dimension 1) are fixed to two distinct digits selected from the set $\{1, 2, \dots, n\}$. As an example, $XX24X$ specifies a S_3 in S_5 , but $3XXX5$ does not. As mentioned in the previous section, each node failure destroys $\binom{n-1}{k} S_{n-k}$'s in S_n . On the other hand, a substar contains a node if its label covers the label of that node. For instance, $XX3X5$ contains 12345 but does not contain 12435. The goal is to find a set of nodes that provide a minimal covering of S_{n-2} 's in S_n , wherein the selected nodes include all 2-digit permutations on n digits in all positions from 2 to n . This will guarantee that all possible S_{n-2} 's will be damaged, should the selected nodes become faulty. For convenience, we first assume n is a prime number and demonstrate our approach using an example.

Example 2 (Derivation of $f(5, 2)$). We select 4 nodes such that the label of the node I is $(1, 1+I, 1+2I, \dots, 1+5I)$ where all additions are modulo 5, and $1 \leq I \leq 4$. Specifically, the set of nodes selected are: $\{(12345), (13524), (14253), (15432)\}$. Each of these nodes is then rotated cyclically to generate 5 distinct permutations. The set of nodes corresponding to the 20 permutations obtained as such optimally damage all S_3 's. The generalization is given below.

To obtain a minimal set of nodes that damage all S_{n-2} 's in S_n :

Step 1. Select n nodes such that the label of the node I is $[1, 1+I, 1+2I, \dots, 1+(n-1)I]$ where all addi-

tions are performed modulo n . Each node I is said to be the leading node of the I th class.

Step 2. Rotate the labels obtained in step 1 cyclically, one digit at a time, to generate a set of $n(n-1)$ labels. These labels specify the set of nodes that destroy all S_{n-2} 's in S_n .

Correctness proof. It suffices to show that all 2-tuple permutations of distinct digits in $\{1, 2, \dots, n\}$ are covered by at least one label in the set of labels generated in step 2 above. For a given 2-tuple $I_L J_M$ (i.e., digits I and J occupying positions L and M), let $(M-L)D = J-I \pmod{n}$. Solving this equation for D , we can easily obtain the label of the node that covers the 2-tuple by cyclically rotating the leading node of class D to left by $(Y+2-L)$ positions where Y is obtained from the equation $YD + D + 1 = I \pmod{n}$. (This is because the class D node leader has its second digit as $D+1$. By virtue of the construction, the current position of I in the class D node leader is Y digits to the right of the second position.)

Example 3. Let the 2-tuple be $(2_2 4_5)$. It follows $(5-2)D = (4-2) \pmod{5}$. This will yield $D = 4$, implying that the covering node belongs to the 4th class of leading nodes (i.e., 154326 as in Example 2). It follows $Y4 + 4 + 1 = 2 \pmod{5}$ which results in $Y = 3$. Rotating (154326) to left by 3 digits will render the node (326154) which covers the said 2-tuple. We present the following theorem.

Theorem 1. Let n be a prime number. The optimal number of nodes to damage all S_{n-2} 's in S_n is

$$f(n, 2) = n(n-1).$$

Proof. By way of construction as discussed before. Optimality follows the fact that there are $n(n-1)$ disjoint S_{n-2} 's in S_n . $\square$

Next, we consider the case where n is not a prime. As before, we clarify the approach using an example.

Example 4. Let $n = 4$. We consider $f(p, 2)$ where p is a prime number such that $p \geq n$. In this case, $p = 5$ and we proceed as if we want to derive the list of faulty nodes for S_5 . However, as digit 5 is not present in S_4 , when forming labels of the leading nodes, we simply use “-” instead of 5 as shown below.

$$\{(1234-), (13-24), (142-3), (1-432)\}.$$

Next, we rotate the leading nodes cyclically one digit at a time, leaving out dashes at the end to obtain the following:

1234, 2341, 3412, 4123,
 1324, 3241, 2413, 4132,
 1423, 4231, 2314, 3142,
 1432, 4321, 3214, 2143.

The set of nodes given above will damage all S_2 's in S_4 . However, the set is not optimal ($f_{\text{optimal}}(4, 2) = 12$). The nonoptimality is a direct consequence of n not being a prime number, which necessitated the formation of $(n + 1)$ leading nodes as opposed to only n where n was prime.

Theorem 2. Let p be a prime number such that $p \geq n$. The number of nodes to damage all S_{n-2} 's in S_n satisfies the following bound:

$$f(n, 2) \leq n(p - 1).$$

Proof. Since we use dimension p as the basis for our node generation, by Theorem 1 all two-tuples of the form $I_L J_M$ are covered such that $\{I_L, J_M\} \in \{1, 2, \dots, n, \dots, p\}$ and $\{L, M\} \in \{2, \dots, n\}$. Clearly any 2-tuple $I'_L J'_M$ where $\{I'_L, J'_M\} \in \{1, 2, \dots, n\}$ will also be covered. There is no danger in losing any 2-tuple due to wrap around adjacency as dashes (which can be thought of as digit n repeated $(p - n + 1)$ times) are all collapsed into n . $\square$

Table 1 indicates the value of $f(n, 2)$ for various dimensions. Dimensions for which there is an exact value are shown in bold.

When n is not prime, one may be tempted to build the faulty set based on a prime number $p' < n$, by adding extra nodes to the existing ones. We considered this approach, but the results obtained were worse than what is given in Theorem 2.

Table 1
 $f(n, 2)$

n	Lower bound	Upper bound
3	6	6
4	12	12
5	20	20
6	30	36
7	42	42
8	56	80
9	72	90
10	90	100
11	110	110
12	132	144
13	156	156

Case (ii). Derivation of $f(n, k)$ for $3 \leq k < n - 2$

A sketch of the method is presented here without proof. To construct the faulty nodes, we follow an extension of the method presented for $k = 2$. The main idea is to first form the node leaders in classes I , $1 \leq I \leq n - 1$, as before. For each class I node leader, form n nodes by cyclically shifting the leader's label one digit at a time. (We would stop here if $k = 2$.) For each node obtained as such, keep the first digit in place, and produce $(n - 1)$ nodes by cyclically shifting the rightmost $(n - 1)$ digits, and stop if $k = 3$. Otherwise, for each node, keep the first two digits in place and generate $(n - 2)$ nodes by cyclically shifting the remaining $(n - 2)$ digits to the right. We would stop here if $k = 4$, otherwise we continue. Finally in the last step, for each node, we produce $(n - 2 + k)$ nodes by keeping the first $(k - 2)$ digits in place and rotating the remaining $(n - 2 + k)$ digits, one digit at a time. The labels obtained as such cover all permutations of length k in an n -tuple, and therefore correspond to the set of faulty nodes that collectively damage all S_{n-k} 's in S_n .

We demonstrate the approach for $n = 5$ and $k = 3$.

The leader nodes are given below:

$$\{(12345), (13524), (14253), (15432)\}.$$

The first shifting of these nodes, say (12345), will result in the following set: $\{(12345), (23451), (34512), (45123), (51234)\}$. Each of these nodes, say (23451), will have their first digits fixed and the remaining digits rotated to generate: $\{(23451), (24513), (25134), (21345)\}$. The 80 nodes obtained as such constitute the faulty set that damages all S_2 's in S_5 . Note that we already know 60 nodes are sufficient to destroy all S_2 's in S_5 , but the example given is for illustrative purposes only.

Conjecture 1. Let p be a prime number such that $p \geq n$. The number of nodes to damage all S_{n-k} 's in S_n for $3 \leq k < n - 2$, satisfies the following bounds:

$$\frac{n!}{(n - k)!} \leq f(n, k) \leq (p - 1) \frac{n!}{(n - k + 1)!}.$$

4. Conclusion

The upper bound on the minimum number of faulty nodes to make every substar of a given size faulty was improved, and in some cases the exact value has been derived. Development of a formal proof for the conjecture presented is still under our investigation. The results can be used in reliability analysis of the subnetworks in the star graph. The problem could be generalized to include embedding of graphs in a given host

graph H . In the generalized form, one may wish to investigate the minimum number of nodes that would render all subgraphs G of certain size in H unusable. In a special case, G can be a ring (or a Hamiltonian cycle if a maximum size ring is desired), mesh, or any other topology. The general problem, though, appears to be difficult since even for a well-defined special case tackled in this paper, only bounds on the minimum number of nodes are known.

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